

DOPA1 Pre-Registration Clarifications v0.7

DOPA1 — PRE-REGISTRATION CLARIFICATIONS

| v0.7 · SIGN-OFF ADDENDUM

Operational definitions to be completed before confirmatory collection

Addendum to: DOPA1 POC Scope & Acceptance **v0.5.1** (frozen) · **Responding to:** §16 Feasibility Classification (Modeller)

*Note on numbering: this addendum's own revision number (**v0.7**) is unrelated to Scope **v0.5.1**. Two separate documents, two separate version series. The Scope is unchanged and remains at v0.5.1. **From:** Gargi (DOPALAND) · **To:** Debanjan*

| 1. Status and governing documents

This is an **operational addendum** to Scope v0.5.1. It is not a replacement scope document and does not restate the frozen architecture. It records only the decisions and operational definitions that must be fixed before confirmatory collection.

Governance order:

1. **Scope v0.5.1** governs scope, architecture and exclusions.
2. **This addendum** governs the D1-D8 operational clarifications and the corrections listed here.
3. **The completed pre-registration** is the signed execution record.
4. Later substantive changes follow the §12 change-control process.

Where this addendum is silent, **v0.5.1 remains authoritative**.

The modeller proposes. Gargi signs off. Nothing here is confirmed merely because it appears in this document.

1.1 Two genuinely new requirement areas — flagged honestly

Everything below traces to v0.5.1 or to your §16 response, **with two exceptions**, both added deliberately and both cheap:

- **§16 Stopping and exclusion rules** — absent from v0.5.1. Added because outcome-dependent exclusion is one of the few remaining routes to an unfalsifiable result.

- **§17 Privacy and retention** — absent from v0.5.1. Added because it must be documented before recordings exist, not after.

These are additions. They are labelled as such rather than presented as if they were always there.

2. D1 — Feature-block separation

2.1 Required evidence

1. A **feature dependency map** for every component of E_t , traced to source variables, reviewable by someone who did not write the code.
2. An **explicit exclusion list** demonstrating that ROI, orientation, dwell, persistence, switching, head-gaze coherence and any downstream attention derivative cannot enter E_t .
3. **Versioned feature manifests for all five blocks** — X_{core} , E_t , A_t , U_t , Z_t .

To be precise about what separation means here: **derived representations may depend on their permitted upstream blocks**. $E_t = h(X_{core}, \Delta X_{core}, C_t)$ by definition, so E_t and X_{core} necessarily share information — that is the architecture, not a defect. The requirement is **directional**: attention-derived information and audio must not flow *into* X_{core} or E_t .

2.2 Both exclusions apply

v0.5.1 §3.1 imposes **two** separation constraints on the core, not one:

PERMITTED FLOW	FORBIDDEN FLOW
$X_{core} \longrightarrow E_t$	$A_t \not\longrightarrow X_{core}$
$C_t \longrightarrow E_t$	$A_t \not\longrightarrow E_t$
	$U_t \not\longrightarrow X_{core}$
	$U_t \not\longrightarrow E_t$

If attention reaches the core, $\Delta_{attention}$ compares a model against itself. If audio reaches the core, Δ_{audio} measures nothing.

The audio constraint is easy to lose because the attention discussion dominates. Please confirm both explicitly in the dependency map.

2.3 Lag diagnostic

$X_{core,t} \rightarrow A_t$

- **Near-perfect at zero lag** → contamination flag; triggers a code/timestamp audit. Not automatic proof of leakage — synchronous physical relationships and shared timestamping can both produce it.
- **Moderate at lag k** → plausibly genuine behavioural coupling; interpret through the dependency map and lag structure.

The architectural test is the dependency map. The diagnostic is subordinate to it.

2.4 Failure consequence

If clean separation cannot be demonstrated, **do not run the invalid comparison**. The §16 feasibility window is closed; the existing §12 change-control process applies, and the comparison is re-specified before it is run.

3. D2 — Prediction target and horizon

3.1 Action space

The class set must be **enumerated from the actual controlled software environment**. The illustrative form in v0.5.1 §7.1 —

$Y_{\{t+1\}} \in \{ A, B, C, ABANDON, NO_ACTION \}$

— is an example of the *shape*, not a specification of the ROI count. ABANDON and NO_ACTION must be real classes, not excluded trials.

3.2 To return

- Proposed prediction window $[\Delta t_{min}, \Delta t_{max}]$ with task-based justification
- Tie handling
- Rapid-successive-action handling
- Whether the target is first action, sequence, or trial outcome when multiple actions occur
- Missing-observation handling, kept distinct from NO_ACTION
- Expected class balance, **derived from pilot data or task design**, stated before confirmatory collection
- If class weighting is used: the method, with weights fitted on **training data only**. No post-hoc resampling or weighting choice.

4. D3 / D7 — Reliability, ICC and baseline

4.1 ICC specification (D3) — and the repeated-unit problem

Return first: what is the repeated unit? This must be settled before any ICC is computed, and it is the one item in this document that should hold up collection if it cannot be answered.

ICC is a variance-components statistic: it estimates the ratio of between-unit variance to total variance. It therefore **requires meaningful between-unit variance to exist**. In a conventional test-retest study the units are subjects, each measured repeatedly.

This POC is deliberately **one person**. So:

NOT A VALID DESIGN

- one summary value per session × 3 sessions
- three observations of a single unit
- no between-unit variance → ICC is uninterpretable

VALID DESIGN

- matched repeated units within each session
- units = scripted episodes / trials; measurements = sessions

Unit	Day 1	Day 2	Day 3
E1	x	x	x
E2	x	x	x
E3	x	x	x
...	...	...	...

If ICC is retained, return:

1. The repeated unit, explicitly named.
2. Evidence the units are **matched and comparable across sessions** — same scripted episode, same stimulus, same position in the protocol.
3. The ICC model, chosen to match that structure (units random, sessions as repeated measurements), with confidence intervals.

Heterogeneity caveat — this cuts the other way and is easy to miss. Because ICC is a ratio, it is depressed when the units are too similar to each other. If all scripted episodes elicit near-identical signal values, ICC will be low **even with a perfect instrument**, and you will wrongly conclude the signal is unreliable. The reliability protocol should therefore include episodes **spanning the range of signal values relevant to the intended use**. State how that range was designed.

To be explicit about the objective, because the instruction inverts easily: this is **representative coverage of the intended operating range**, not maximisation of heterogeneity to make ICC look better. Designing artificially extreme episodes to inflate the ratio would be exactly the wrong response.

If no valid repeated-unit structure exists, do not report an ICC. Use absolute reliability measures instead, which do not depend on between-unit heterogeneity:

- **Standard error of measurement (SEM)** — within-unit measurement spread, in the signal's own units
- **Repeatability coefficient** — $RC \approx 1.96 \times \sqrt{2} \times SEM$; a change larger than RC is unlikely to be measurement noise
- **Bland-Altman limits of agreement** between session pairs
- **Within-unit coefficient of variation**

Worth noting these may in fact suit DOPA1 better than ICC: the deployment claim — “*is today unusual for this person?*” — is precisely a question about whether an observed change exceeds measurement noise, which is what the repeatability coefficient answers directly and ICC does not. Propose which family you will use and why.

4.2 Three representations (D7) — all three are computed

Design: three sessions in total, conducted on three separate days — not three sessions per day. The temporal separation is the point; the whole D7 question is what happens to the reference frame *between* days.

Those three sessions **are analysed under all three representations, on the same recordings:**

ICC_raw	raw signals
ICC_session-z	$z = (x_t - \text{median}_s) / (1.4826 \cdot \text{MAD}_s)$
ICC_persistent-z	$z = (x_t - B_{\text{person},t}) / S_{\text{person},t}$

This is not conditional. What is *not* required is building a full longitudinal deployment baseline system. What **is** required is defining a persistent-baseline update rule sufficient to compute ICC_persistent-z on these three sessions, and computing it.

Define the scale $S_{\text{person},t}$ — it is currently unspecified and it carries the same leakage risk as the baseline.

$S_{\text{person},t}$ is the person-level dispersion used to standardise. Return: the **estimator**, the **update/window rule**, and the **temporal rule**. Candidate estimators include historical SD, robust scale via $1.4826 \times \text{MAD}$, pooled within-person variability, a rolling historical scale, or scale from the immediately preceding session — pick one and justify it.

The temporal discipline applies to the scale exactly as it does to the baseline:

$B_{\text{person},t} = f(x_{\{1:t-1\}})$	and	$S_{\text{person},t} = g(x_{\{1:t-1\}})$	CORRECT
$B_{\text{person},t} = f(x_{\{1:t-1\}})$	but	$S_{\text{person},t} = g(x_{\{1:T\}})$	LEAKAGE
paired with a			— a correct baseline
			scale fitted on

all data, including

the future

A prospectively-computed numerator divided by a retrospectively-computed denominator is not a prospective representation. $S_{\text{person},t}$ must be estimated **only from observations available before the observation being evaluated**.

Temporal rule — the baseline must not see the session it is evaluating.

$B_{\text{person},t} = f(x_{\{1:t-1\}})$

CORRECT — history strictly before t

$B_{\text{person},t} = f(x_{\{1:t\}})$

LEAKAGE — the session updates its own baseline

A baseline estimated using the very session being evaluated is not a persistent baseline; it is a within-session baseline wearing a different name, and $ICC_{\text{persistent-}z}$ will be optimistically biased. Return the update rule for **both $B_{\text{person},t}$ and $S_{\text{person},t}$** and state explicitly **whether the current session contributes to its own baseline or its own scale**. For any prospective interpretation, current and future observations must not update the baseline before the observation being evaluated.

Consequence for a three-session study — state this in the report. Under the correct rule, session 1 has no prior history and therefore no persistent baseline; the representation is computable only from session 2 onward. $ICC_{\text{persistent-}z}$ will therefore rest on fewer session-pairs than the other two representations. That is expected, not a defect — but the three ICCs are then not computed on identical amounts of data, and the comparison must say so rather than presenting them as like-for-like. Also state whether the baseline **and the scale** use an expanding window over all prior sessions or the immediately preceding session only. They need not use the same window, but each must be stated.

4.3 Claim → representation mapping — pre-specified

Claim	Representation
POC — predict this person's on-screen action within a controlled session	within-session calibrated
Deployment — determine whether today is unusual relative to this person's history	persistent cross-session baseline

All three ICCs are reported side by side. The mapping is fixed before the results are inspected. Any later change is an explicit amendment under §12, labelled as such — not a silent substitution.

Rationale, for the record: per-person **within-session** calibration and a **cross-session** reliability study do not automatically compose, because the reference frame moves between sessions. Fixing the mapping in advance prevents the POC's answer being borrowed later for a claim it never tested.

4.4 Timing

Reliability sessions run **before** confirmatory collection, so an unstable signal is identified before it becomes a core model input.

5. Decision rules — the acceptance criteria

This section restores material from v0.5.1 §§7.3, 7.5 and 8 that must be operationally defined and was not carried into earlier drafts of this addendum.

5.1 Primary and secondary metrics — and metric orientation

Return **one primary metric** appropriate to the final target type and class structure, plus secondary metrics.

Orientation must be fixed first, because the permitted metrics do not all point the same way. Balanced accuracy and macro-F1 are higher-is-better; log loss and calibration error are lower-is-better. Every decision rule below is written as $\Delta > 0$ means improvement, which silently inverts if a lower-is-better metric is used.

Define an **oriented utility** U , such that higher U is always better:

$U = \text{macro-F1}$ metric)	or	balanced accuracy	(higher-is-better metrics: $U =$
$U = -\log \text{ loss}$ - metric)	or	- calibration error	(lower-is-better metrics: $U =$
$\Delta = U(M_1) - U(M_0)$		$\Delta > 0 \leftrightarrow$	improvement, for every metric

State U explicitly in the pre-registration. **Every** minimum-effect threshold below is then expressed **in the units of U** — and if the primary metric is ever changed under §12, all of them must be re-derived, not carried across. This removes a hidden fork in which the sign of the comparison could be chosen after seeing the result.

Threshold nomenclature — four separate thresholds, four separate decisions.

They are named distinctly on purpose:

δ_{Gate3} (§5.3)	minimum practically meaningful improvement for M_{core} vs M_{0b}
$\delta_{\text{attention}}$ (§5.4)	minimum practically meaningful contribution for $\Delta_{\text{attention}}$
δ_{audio} (§5.4)	minimum practically meaningful contribution for Δ_{audio}
δ_{latent} (§5.4)	minimum practically meaningful contribution for Δ_{latent}

There is no generic δ_{min} in this document. Each threshold is proposed and justified **separately**, because the four decisions are different: beating a trivial baseline is not the

same question as a component earning its place on top of an already-working core, and the three components differ from each other in cost and expected effect. They may coincide numerically — but only if you state that they do and why, never by default because they share a name.

- If AUC is used on a multiclass target, state **one-vs-rest or one-vs-one**, and **macro or weighted** averaging. “AUC” alone is not a specification.
- Balanced accuracy, macro-F1, multiclass log loss and calibration error are acceptable candidates.
- Accuracy alone is insufficient where class imbalance is material.
- Justify the choice in one line against the class structure.

5.2 Uncertainty and resampling unit

- Report confidence or bootstrap intervals on every primary comparison.
- **Resample at episode / session / day level. Never bootstrap individual frames or trials as if independent** — behavioural data is autocorrelated and frame-level resampling will badly understate variance.
- For every incremental comparison, report the uncertainty **of the difference itself**, not of the two components separately.
- **Permutation test — specify the permutation unit.** These are serially dependent behavioural observations. Shuffling individual trials destroys the temporal structure and violates exchangeability, producing a null distribution that is too narrow and a p-value that is too small. Permutation must **preserve the relevant dependence structure**: pre-specify the **permutation scheme and exchangeability unit** appropriate to the primary hypothesis (episode / session / day). The permutation scheme **need not be identical to the bootstrap resampling scheme** — bootstrap estimates the uncertainty of an estimator, permutation constructs a null distribution under a specific hypothesis, and the appropriate unit can legitimately differ. Both must respect the temporal dependence structure. ≥ 1000 permutations unless computationally infeasible, in which case state the number and why.

5.3 Gate 3 — the M_{core} vs M_{0b} rule

Return the decision rule in advance. Required shape:

```
M_core beats baseline ⇔ LB[ U(M_core) - U(M0b) ] > δ_Gate3

                                LB = lower bound of the CI on the difference in ORIENTED
UTILITY U
                                δ_Gate3 is expressed in units of U (§5.1)
```

M_{core} beats the baseline if and only if the lower bound of the confidence interval on $U(M_{core}) - U(M_{0b})$ exceeds δ_{Gate3} , where **U is the oriented utility defined in §5.1** and δ_{Gate3} is expressed in units of U. The raw metric is never differenced directly, because the sign would invert for lower-is-better metrics.

δ_{Gate3} — the smallest difference that would be practically meaningful **for this comparison** — is proposed and justified by you, and is distinct from the component thresholds in §5.4. Both outcomes are acceptable POC results. The only unacceptable outcome is an uninterpretable one.

5.4 RETAIN / DROP / INCONCLUSIVE — the three-way verdict

The POC terminates in this verdict for each of $\Delta_{\text{attention}}$, Δ_{audio} , Δ_{latent} . It must be a **rule applied**, not a judgement made at analysis time. Required shape:

For each component $c \in \{ \text{attention, audio, latent} \}$, using its own threshold δ_c :

RETAIN	lower bound of CI on $\Delta_c > \delta_c$	evidence of benefit
DROP	upper bound of CI on $\Delta_c < \delta_c$	evidence of ABSENCE of meaningful benefit
INCONCLUSIVE	CI spans δ_c	underpowered / cannot support a decision

Return **$\delta_{\text{attention}}$, δ_{audio} and δ_{latent} separately**, each with its own justification, each in units of U. Do not carry δ_{Gate3} across — it governs a different comparison. If two thresholds are set to the same value, say so explicitly and give the reason; equality must be a decision, not an artefact of shared notation.

DROP and INCONCLUSIVE are not interchangeable. DROP is a positive finding — the component was tested with enough precision to rule out a meaningful contribution. INCONCLUSIVE means the test could not decide. Reporting the second as the first overstates the result.

5.5 Multiple comparisons and exploratory status

Distinguish pre-specified primary outcomes from exploratory analyses in the report. An exploratory positive is not promoted to confirmatory evidence after the fact. Δ_{latent} is exploratory by construction (v0.5.1 §8.3) — a positive result is labelled exploratory and cannot on its own justify the funded phase.

5.6 No post-hoc threshold movement

Every threshold governing a pass/fail or retention decision is fixed before the relevant test. A changed threshold is an amendment under §12 and the affected analysis is relabelled amended or exploratory.

| 6. Data integrity and temporal leakage

Also restored from v0.5.1 §5.4 — absent from earlier drafts of this addendum.

6.1 Temporal rule

Features used to predict at time t must be computable from information available by t . No post-action frames, post-action audio, future interaction logs or future-derived state may enter the predictive window.

6.2 Preprocessing and split hygiene

- Normalisation and any other learned preprocessing parameters are fitted on **training data only** — never across the full dataset before splitting.
- Model selection and threshold selection must not touch the prospective test set.
- Chronological splits only; no future session or trial informs training.
- **Nested, temporally ordered selection.** Any hyperparameter, feature-selection or model-family choice must be made using **training data only**, with validation folds or splits that respect temporal order, arranged chronologically as $\text{train} < \text{validation} < \text{test}$. The prospective test set is opened **once**, for the frozen evaluation. Without this, one can technically honour “don’t touch the test set” while optimising repeatedly against the whole training set through unrecorded internal choices — and ordinary random cross-validation on time-series data leaks by construction.

6.3 Leakage controls — to be operationally defined and run

```
VALID WINDOW
past ----- t_predict ----- ACTION
           [features strictly before action]

POST-ACTION CONTROL (deliberately invalid)
past ----- ACTION ---- [window includes post-action information]

PRE-ACTION CONTROL (deliberately starved)
past ---- [window ends well before] ----- ACTION

TIMESTAMP-SHIFT CONTROL
shift action timestamps back ~2 s and re-run
```

Return, before running: what counts as a **substantial** difference **for this task**, and the reporting format.

Task-conditional caveat. These establish expected *direction*, not magnitude. Where post-action information is not known to carry target-relevant signal for this task, the post-action window may not dramatically outperform, and that alone does not prove the windowing is broken. If the controls do not show the expected direction, **investigate the windowing implementation before interpreting any predictive result**, and report the outcome of that investigation either way. Diagnostics that must be run and reported — not automatic kill gates.

6.4 Timestamp integrity

Confirm: synchronisation across webcam, audio and software events; monotonic frame timestamps; logged event times matching real screen events; ROI coordinates matching real screen geometry; correct person_id / session_id throughout; missingness logged rather than dropped.

| 7. D4 — Reproducibility

7.1 Same pinned environment

Same code, dependencies, data, configuration and seed → **deterministic replay expected**. Any discrepancy is investigated as an engineering defect (unseeded randomness, threading, ordering), not accepted as a floating-point fact.

7.2 Different environment

Across machines, **reproducibility-to-tolerance** is acceptable. Return the tolerance and the comparison method.

7.3 The named executor

The acceptance test is: **Gargi runs the documented reproduction command on a clean machine and the reported results tables regenerate**. A repository and a README do not satisfy this. This is simultaneously the reproducibility control and the point at which the deliverable becomes usable independently of its author.

7.4 Required evidence

Pinned dependency/environment specification · configuration and version hash · Experiment ID and commit · seeds where stochasticity exists · a single documented reproduction command · numerical comparison or checksum evidence.

| 8. D5 — Actual action

Off-screen / lived-environment “actual action” remains a **named funded-phase open item**. The POC’s confirmatory target is objectively logged on-screen action. Nothing in the report may imply that this validates real-world action capture.

| 9. D6 — Precision simulation

The sequence is mandatory and not reorderable:

SIMULATE → required information / N → feasibility decision → COLLECT

- Do **not** assume IID trials. Model serial dependence, and include learning, fatigue, class-frequency drift and expected missingness where material.
- State the target effect or precision, primary metric, train/test structure, dependence assumptions and stopping rule.
- The artefact is dated and versioned, submitted **before** confirmatory collection.
- **Escape hatch:** if the required N is infeasible within POC constraints, say so at that point and the claim is scoped down before collection. An underpowered negative result is not a finding.
- No fixed N is imposed by this addendum.

Efficiency note: the synthetic generator required for §10.7 can produce the trial sequences needed here. Build once, use twice.

| 10. Controls

10.1 Positive control — blink (restored; needs a pass criterion)

Ten one-minute clips, frame-by-frame manual blink counts, compared against detector output.

Return **before running**: the event-matching tolerance, the metric (event precision/recall, or count agreement), **and the pass criterion**. A positive control with no threshold cannot fail, and a control that cannot fail is not a control.

This validates blink-count detection. It does not validate any psychological interpretation of blinking.

10.2 Null-input control

Person still, blank screen, ten minutes, full pipeline. Return **before running**: the event/excursion rule, the false-event-rate metric, and the acceptable threshold with its justification. A natural non-arbitrary reference: the null-input false-event rate should not exceed what the negative control produces under the same rule.

10.3 Negative control

A deliberately meaningless signal carried through **every** test alongside the real ones. It must not demonstrate contribution beyond the pre-specified null distribution; if it does, the corresponding analysis is treated as compromised and investigated before the real signal is interpreted. Not an automatic kill rule — across many tests a negative control will occasionally reach significance by chance.

10.4 Time-shuffle

Shuffle episode order and re-run. If ordered and shuffled performance are comparable, temporal organisation is contributing little — a direct check on the behavioural-dynamics claim. Return the reporting format.

Diagnostic only. Do not interpret the resulting difference as a formal permutation-test p-value; the inferential permutation procedure with its pre-specified exchangeability unit is defined separately in §5.2.

10.5 Modality ablation

Required — the primary test: M_{core} vs $M_{\text{core}} + U_t$, at test time. This is how Δ_{audio} is established, and it is the question v0.5.1 actually asks: *does audio add predictive information beyond the core model?*

Optional diagnostic: audio-only performance may be reported if you consider it informative, but it is **not required** and does not feed the POC verdict. It answers a different question — *can audio independently predict the action?* — which becomes relevant later for coverage-driven degradation, when vision is unavailable. Not now. Do not build an analysis branch for it merely because it is mentioned here.

10.6 Context stress test

Two sessions differing in lighting and camera placement. Question: does the result obviously disappear?

This is a robustness / stress test only. It is not interpreted causally. Lighting and camera placement change together, and camera placement is also a sensor/instrument variable already covered by §10.8 — so a drop cannot be attributed to lighting, to geometry, or to either individually. Report it as “*performance under a perturbed recording setup*”, nothing more.

If the stress test does show a drop, the sensor-swap result (§10.8) helps disambiguate: swap isolates the instrument with behaviour held constant, so an instrument effect there points away from lighting here. The two are complementary diagnostics.

Neither is the full context-envelope programme, which remains funded-phase. Do not add further experimental arms to decompose this.

10.7 Synthetic latent recovery — prerequisite for Δ_{latent}

$Z_{\text{true}} \rightarrow \text{observation generator} \rightarrow \text{pipeline} \rightarrow \hat{Z}$ recovery error = $d(Z_{\text{true}}, \hat{Z})$

Run before M2 touches human data. Return the recovery-error metric and the threshold that constitutes successful recovery.

Without this, $\Delta_{\text{latent}} \approx 0$ is ambiguous between “*the latent state adds nothing*” and “*the implementation is silently broken*” — which destroys the value of the negative result.

Synthetic recovery validates the **machinery** under known assumptions; it establishes nothing about whether a human latent state is psychologically real.

10.8 Sensor swap

Two cameras recording simultaneously during the same behavioural session. Sequential re-recording is **not a substitute** — it varies behaviour and time as well as the instrument, so any difference is unattributable. Free if done during collection; permanently impossible afterwards. If omitted, the omission is documented explicitly rather than silently.

10.9 Audio — acquisition versus analysis

If Δ_{audio} is to be evaluated, **microphone capture and time-synchronisation must be in place before confirmatory collection**. The analysis may return DROP without failing the POC; absent acquisition makes the analysis impossible forever. Record microphone availability, audio quality and missingness. Document the clock-synchronisation method.

| 11. D8 — Attention validity

Before Gate 2 is run, return for sign-off: **primary statistic · null distribution · minimum practically meaningful effect · required trial count · uncertainty interval · failure consequence**.

Gate 2 logic:

Salience	→	Orientation
Non-choice salience	→	Orientation without Choice

Test C uses a salient ROI **outside the choice set**; the participant is not instructed to ignore anything. Failure to detect orientation at the non-choice ROI indicates the measure is recovering choice rather than measuring orientation.

Failure rule. If the pre-specified criterion fails, attention features are removed from the predictive model, M_{core} runs without them, and the report states explicitly: *“Behavioural prediction was evaluated without a validated attention measure.”* Declared before Gate 3, not discovered after. Attention failure does not fail the POC.

| 12. Scope corrections

12.1 LLM behavioural / personality read

Listed as implemented in your §16 Part 1, but absent from v0.5.1 in both the in-scope and exclusion tables. That is a gap in my document. Correcting it:

Status: demonstration and interface only. Not a validated measurement, not a POC outcome, not evidence for V_{es}/V_{pd} or any pleasure/pain inference.

Features → model → LLM interpretation → TERMINAL OUTPUT

Output may be logged, displayed and qualitatively inspected. It must **never** feed back into feature extraction, calibration, state estimation, prediction, target construction or model updating. Demo language must preserve the underlying epistemic status: a candidate pleasure-side signal is not described as validated pleasure, and no pain inference is presented as established.

12.2 V_{es} / V_{pd} and candidate signals

Use the wording: *“implemented; pilot-scale candidate signals; not validated as psychological constructs.”*

Brow/jaw remain logged-only with no validated pain axis. Blink and gaze remain experimental/unvalidated, except for whatever detector performance the positive blink control establishes — and detector validation is not construct validation.

13. A-column verification

You proposed this yourself: that the Part 1 claims be re-checked against committed code and Experiment ID before the response is authoritative. Agreed.

The table below is transcribed from **your §16 Part 1** (the eleven A rows). Return the two right-hand columns. Any item that cannot produce a commit/Experiment ID and an artefact is **reclassified B** until verified.

#	A item (from your §16 Part 1)	Commit / Experiment ID	Artefact / evidence
1	Existing webcam capture pipeline, real-time (two-thread)		
2	Face / head / visible-torso tracking (FaceMesh + Pose)		
3	Sustained ≥15 FPS with full processing		
4	Candidate facial signals (geometric vectors from landmarks)		
5			

#	A item (from your §16 Part 1)	Commit / Experiment ID	Artefact / evidence
	Per-person within-session calibration		
6	Rolling-window statistics (avg / peak / variance)		
7	Blink detection — A* experimental / unvalidated		
8	Coarse L/R/centre gaze orientation — A* experimental / unvalidated		
9	Consent + opt-out step		
10	Personality/agent read + full request/response logging		
11	Multi-hour stability soak (no leak / decay / deadlock)		

Please correct any row I have transcribed wrongly. Run this **alongside D1**, not after.

14. Repository, provenance and data integrity

Version control is already implied by v0.5.1 §4 (commit hashes, Experiment IDs, make reproduce). Making the provenance requirements explicit:

- **Raw or identifying data must never enter Git history.** Git history is effectively permanent; a repository that has ever contained face or audio recordings is compromised for its whole life. `.gitignore` from the first commit.
- **Commit a data manifest instead:** file and checksum, subject/session/trial/experiment identifiers, acquisition timestamps, processing version.
- **Raw data lives outside the repository** in the agreed controlled location, access-controlled.
- **Commit the pre-registration to the repository and push it before collection begins.** This is the cheapest available timestamp, and every subsequent amendment appears as a dated diff — which satisfies §12 more robustly than a dated file. Amendments as commits, never as edits.

- **Tag the frozen confirmatory configuration** before prospective collection. The Gate 5 lock needs a named release, not “the latest commit.”
- **Held-out evaluation data is stored separately** from the working repository, with access arranged so that it is not consulted before final analysis. If a genuine separation is impractical for a single developer, say so and we record the limitation rather than claiming a control that does not exist.
- **Variant log** — every materially attempted model, feature set, window or split, logged from the first commit and reported with its count. This is the mechanism that distinguishes a result from a search.

Ownership and access arrangements already agreed remain unchanged; this section concerns provenance and reproducibility, not IP.

| 15. Canonical log and data manifest

Minimum fields sufficient to reconstruct any trial:

```
subject_id · session_id · trial_id · experiment_id · context_id · device_id
commit / config version · model version
stimulus ID and onset · ROI / condition
prediction timestamp · action timestamp · action class
signal · value · confidence
modality availability / quality · missingness flag · missingness reason
prediction output and confidence · ground-truth outcome
```

context_id and device_id are required by v0.5.1 §4 and must not be dropped — C_t is a core model input. person_id / subject_id appears on every observation from the first record, even with one subject.

Return the exact schema, versioned, before confirmatory collection.

| 16. Stopping and exclusion rules — NEW

Not in v0.5.1. Added because outcome-dependent exclusion is one of the few remaining routes to an unfalsifiable result.

Return explicit, pre-specified rules for: **collection stopping · invalid trial · corrupted recording · device failure and whether the session is repeated · aborted session · missing modality · trial and session exclusion · whether repeated trials from a session remain eligible.**

Exclusions must be rule-based and applied independently of outcome. **Trials are not removed because they make the result look worse.**

| 17. Privacy and retention — NEW

Not in v0.5.1. Added because it must be documented before recordings exist.

Return the **actual** storage location, access control, retention period, subject/session pseudonymisation scheme, raw video and audio retention rule, and the feature-extraction and deletion rule where raw data is discarded.

Do not claim privacy properties that have not been implemented. If waveforms are discarded after feature extraction, document the transformation and the deletion rule so the claim is verifiable.

| 18. Change control

Scope v0.5.1 is frozen; this addendum is the operational layer. Any substantive post-sign-off change to the prediction target, feature definitions, baseline mapping, primary metric, any threshold, sample-size logic, inclusion/exclusion rules, model family or acceptance criterion requires: **timestamp · reason · old definition · new definition · impact assessment · version increment · label as pre-specified / amended / exploratory.**

No silent substitutions. Nothing here forbids changing a decision; it forbids changing one invisibly.

| 19. Sign-off matrix

The single response table for this addendum. Every item in the body appears here; nothing falls between two checklists.

#	Item	Operational definition / evidence to return	Status	Sign-off
1	D1 Feature separation	Dependency map · exclusion list · manifests for all 5 blocks · directional attention AND audio exclusions confirmed · lag diagnostic	OPEN	
2	D2 Prediction target	Enumerated classes incl. ABANDON/ NO_ACTION ·	OPEN	

#	Item	Operational definition / evidence to return	Status	Sign-off
		horizon · ties · rapid succession · missingness · class balance · weighting method		
3	D3 Reliability	Repeated unit named + comparability evidence + heterogeneity range · ICC model + CIs, or SEM / repeatability-coefficient alternative with justification	OPEN	
4	D7 Baseline	All three representations computed · update rules for B_{person,t} AND S_{person,t} incl. estimator, window and temporal rule (no self-update, no future-fitted scale) · claim mapping pre-specified	OPEN	
5	Primary metric	Primary + secondary · multiclass convention · oriented utility U so $\Delta > 0$ = improvement · all thresholds in units of U	OPEN	
6	Uncertainty	Bootstrap resampling	OPEN	

#	Item	Operational definition / evidence to return	Status	Sign-off
		scheme · CI on the difference · permutation scheme + exchangeability unit (need not match bootstrap; both respect dependence)		
7	Gate 3 rule	δ_{Gate3} in units of U + decision rule on $U(M_{\text{core}}) - U(M_{\text{ob}})$	OPEN	
8	Δ verdict rule	$\delta_{\text{attention}}$, δ_{audio}, δ_{latent} — separately justified, in units of U · RETAIN / DROP / INCONCLUSIVE rule	OPEN	
9	Leakage controls	Post-action · pre-action · timestamp-shift · what counts as substantial for this task	OPEN	
10	Split hygiene	Preprocessing fitted on training only · nested temporal selection, train < validation < test · test set opened once	OPEN	
11	D4 Reproducibility	Same-env deterministic · cross-env tolerance · reproduction	OPEN	

#	Item	Operational definition / evidence to return	Status	Sign-off
		command run by Gargi on a clean machine		
12	D5 Actual action	Named funded-phase open item + report wording	OPEN	
13	D6 Precision simulation	Simulation + dependence assumptions + feasibility + stopping rule, before collection	OPEN	
14	D8 Attention validity	Statistic · null · minimum effect · N · uncertainty · failure rule	OPEN	
15	Positive control	10 clips · matching tolerance · metric · pass criterion	OPEN	
16	Null-input control	Event rule · false-event-rate metric · threshold + justification	OPEN	
17	Negative control	Control signal · null distribution · carried through every test	OPEN	
18	Time-shuffle	Protocol and reporting format	OPEN	
19	Modality ablation	Required: M_core vs M_core+U_t. Audio-only optional, not part of the verdict	OPEN	

#	Item	Operational definition / evidence to return	Status	Sign-off
20	Context stress test	Two sessions, lighting + placement · reported non-causally	OPEN	
21	Synthetic recovery	Generator · recovery-error metric · success threshold · run before M2	OPEN	
22	Sensor swap	Two simultaneous cameras, or documented omission	OPEN	
23	Audio acquisition	Mic + synchronisation before collection, or Δ _audio formally removed	OPEN	
24	LLM read	Terminal / demo-only · no feedback into any model input	OPEN	
25	V_es / V_pd wording	Candidate / pilot-scale / not validated	OPEN	
26	A verification	11-item evidence table (§13)	OPEN	
27	Repository / provenance	No raw data in history · manifest · tagged freeze · pre-reg committed · variant log · held-out storage	OPEN	
28	Canonical schema		OPEN	

#	Item	Operational definition / evidence to return	Status	Sign-off
		Versioned, incl. context_id and device_id		
29	Stopping / exclusions	Pre-specified rules	OPEN	
30	Privacy / retention	Actual storage · access · retention · deletion rules	OPEN	

20. Execution sequence

1. Gate 0 infrastructure: version control · Experiment IDs · config hash
· pinned deps · variant log · canonical schema
↓
2. NULL-INPUT + NEGATIVE CONTROL on the existing pipeline
(weeks 1–2, before the ROI build – these can invalidate everything downstream)
↓
3. A-claim verification + D1 dependency map
↓
4. Return D2–D8 operational definitions + this addendum's matrix
↓
5. SIGN-OFF · pre-registration committed and pushed
↓
6. Acquire and synchronise microphone + second camera (if retained)
↓
7. Reliability: THREE sessions on THREE separate days → three-way ICC comparison
↓
8. Timestamp-integrity + leakage controls on pilot data
↓
9. Precision simulation → feasibility decision → confirm collection design
↓
10. Synthetic latent recovery (before M2 touches human data)
↓
11. Freeze commit + config + repository tag + manifest
↓
12. CONFIRMATORY COLLECTION (sensor swap runs during; audio captured)
↓
13. M0 / M0b / M_core → $\Delta_{\text{attention}}$ · Δ_{audio} · Δ_{latent}
↓
14. Time-shuffle · modality ablation · context perturbation
↓
15. Prospective frozen test

↓

16. Report RETAIN / DROP / INCONCLUSIVE with uncertainty + failure catalogue

Indicative timing: 12-14 weeks, subject to the §9 precision analysis, which may change the collection design. The calendar is fixed after that artefact, not before.

One commitment in exchange: if the schedule slips, **the controls are the last thing cut, not the first**. A POC delivered on time without its controls proves nothing.

| 21. Final instruction

Return the completed operational definitions and evidence plan against the **single sign-off matrix in §19**.

This is not a request for another scope review. Scope v0.5.1 remains frozen. The purpose of this document is to make the frozen scope executable, auditable and interpretable before confirmatory collection.

The objective is not to make every candidate representation succeed. It is to obtain a clean answer about which representations carry predictive information about this individual's subsequent behaviour, under the conditions actually tested — with the uncertainty stated, and with the failure modes written down.

DOPA1 — Pre-Registration Clarifications v0.7 · Sign-Off Addendum · Addendum to Scope v0.5.1 (frozen)